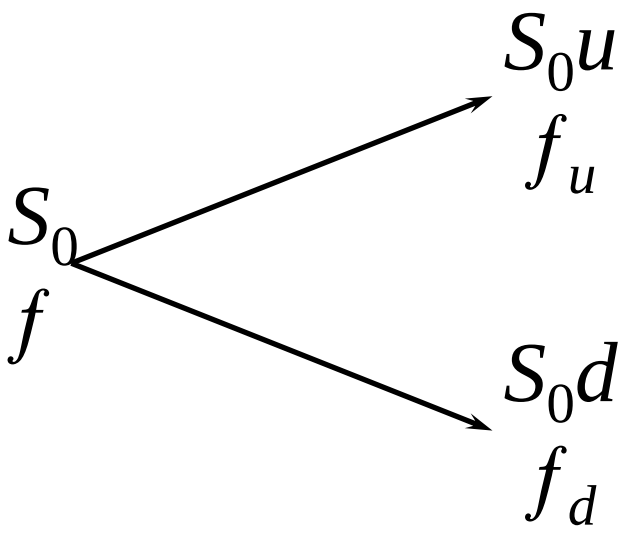
# DERIVATIVES VALUATION

- The Binomial Asset Pricing model provides a powerful tool to understand arbitrage pricing theory and its relation to risk-neutral pricing.
- The underlying assumption is that the asset follows a random walk. In each time step, the asset has a certain probability of moving up by a certain percentage amount and a certain probability of moving down by a certain percentage amount.
- In the limit, as the time step becomes smaller, this model leads to the lognormal assumption for stock prices that underlies the Black-Scholes model.

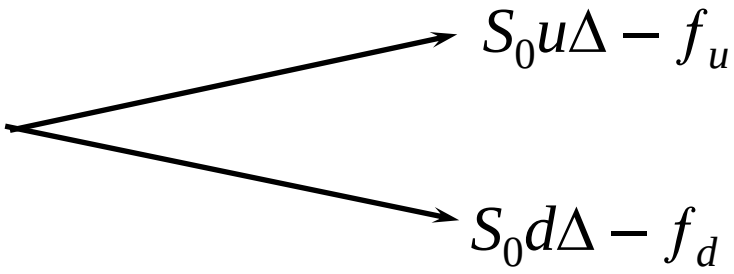
# A SIMPLE BINOMIAL MODEL

A derivative lasts for time $T$ and is dependent on a stock



# GENERALIZATION

Value of a portfolio that is long $\Delta$ shares and short 1 derivative:




The portfolio is riskless when $S_0u\Delta - f_u = S_0d\Delta - f_d$ or

$$\Delta = \frac{f_u - f_d}{S_0u - S_0d}$$

# GENERALIZATION (CONTINUED)

Value of the portfolio at time $T$ is $S_0 u\Delta - f_u$

Value of the portfolio today is $(S_0 u\Delta - f_u)e^{-rT}$

Another expression for the portfolio value today is $S_0\Delta - f$

Hence

$$f = S_0\Delta - (S_0 u\Delta - f_u)e^{-rT}$$

Substituting for $\Delta$ we obtain

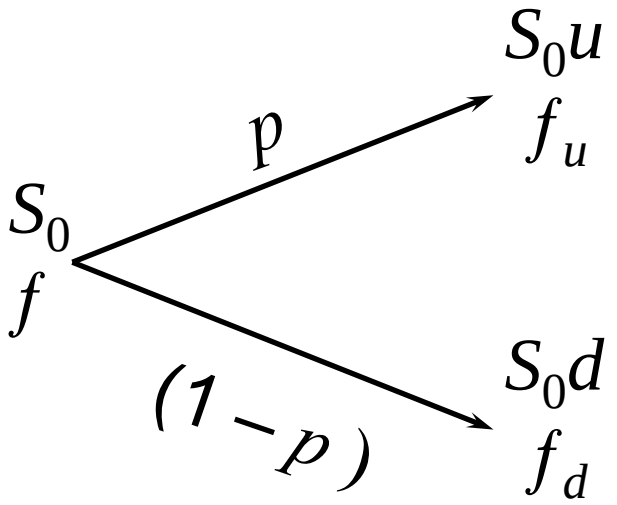
$$f = [\, pf_u + (1-p)f_d \,]e^{-rT}$$

where $$p = \frac{e^{rT} - d}{u - d}$$

# ‘P’ AS A PROBABILITY

It is natural to interpret $p$ and 1-$p$ as probabilities of up and down movements

The value of a derivative is then its expected payoff in a risk-neutral world discounted at the risk-free rate



# RISK-NEUTRAL VALUATION

When the probability of an up and down movements are $p$ and 1-$p$ the expected stock price at time $T$ is $S_0 e^{rT}$

This shows that the stock price earns the risk-free rate

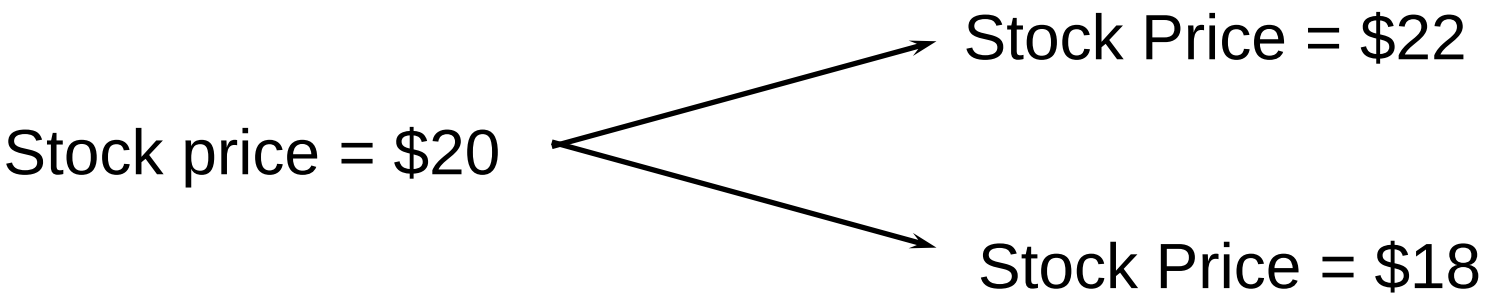
Binomial trees illustrate the general result that to value a derivative we can assume that the expected return on the underlying asset is the risk-free rate and discount at the risk-free rate

This is known as using risk-neutral valuation

# AN EXAMPLE

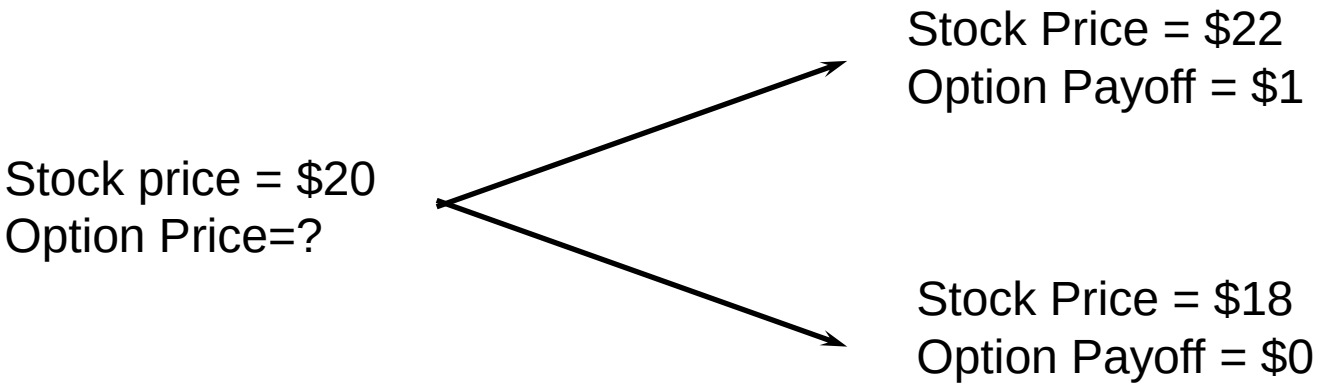
A stock price is currently $20

In 3 months it will be either $22 or $18



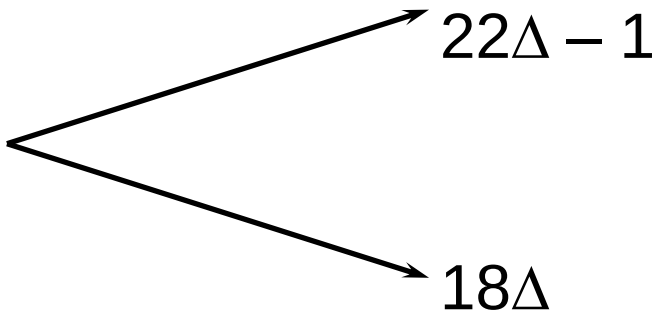
# A CALL OPTION

A 3-month call option on the stock has a strike price of 21.



# SETTING UP A RISKLESS PORTFOLIO

For a portfolio that is long $\Delta$ shares and a short 1 call option values are




Portfolio is riskless when $22\Delta - 1 = 18\Delta$ or $\Delta = 0.25$

# VALUING THE PORTFOLIO
## (RISK-FREE RATE IS 4%)

The riskless portfolio is:

long 0.25 shares

short 1 call option

The value of the portfolio in 3 months is

$22 \times 0.25 - 1 = 4.50$

The value of the portfolio today is

$4.5e^{-0.04\times 0.25} = 4.455$

# VALUING THE OPTION

The portfolio that is

long 0.25 shares

short 1 option

is worth 4.455

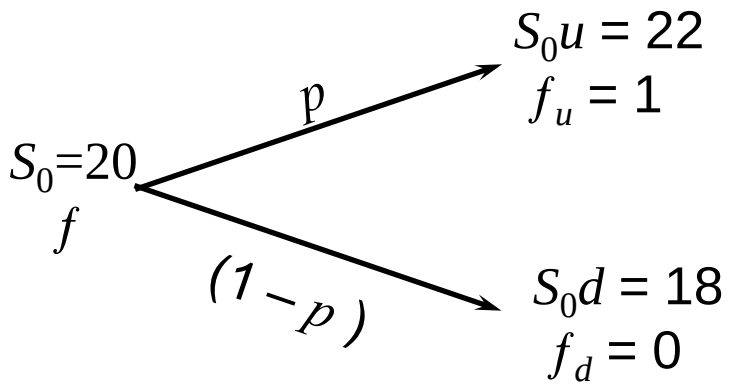
The value of the shares is

5.000 (= 0.25 × 20 )

The value of the option is therefore

5.000 – 4.455 = 0.545

# ORIGINAL EXAMPLE REVISITED



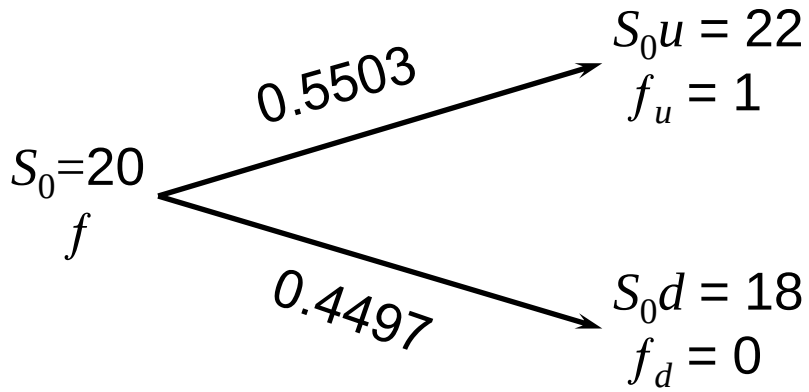

$p$ is the probability that gives a return on the stock equal to the risk-free rate:

$20e^{0.04\times0.25} = 22p + 18(1-p)$ so that $p = 0.5503$

Alternatively:

$$p = \frac{e^{rT} - d}{u - d} = \frac{e^{0.04\times0.25} - 0.9}{1.1 - 0.9} = 0.5503$$

# VALUING THE OPTION USING RISK-NEUTRAL VALUATION




The value of the option is

$e^{-0.04\times0.25}(0.5503\times1 + 0.4497\times0)$

$= 0.545$

# IRRELEVANCE OF STOCK'S EXPECTED RETURN

When we are valuing an option in terms of the price of the underlying asset, the probability of up and down movements in the real world are irrelevant

This is an example of a more general result stating that the expected return on the underlying asset in the real world is irrelevant

Therefore, we could move from a world with one set of risk preferences to another (risk-neutral) world with another set of risk preferences. This process is called 'changing the measure'.

# DELTA

Delta ($\Delta$) is the ratio of the change in the price of a stock option to the change in the price of the underlying stock

The value of $\Delta$ varies from node to node

i.e the Portfolio needs to be rebalanced at each time step in order to stay risk neutral.

# CHOOSING U AND D

One way of matching the volatility is to set

$$u = e^{\sigma\sqrt{\Delta t}}$$

$$d = 1/u = e^{-\sigma\sqrt{\Delta t}}$$

where $\sigma$ is the volatility and $\Delta t$ is the length of the time step. This is the approach used by Cox, Ross, and Rubinstein

# GIRSANOV'S THEOREM

Volatility is the same in the real world and the risk-neutral world

We can therefore measure volatility in the real world and use it to build a tree for the an asset in the risk-neutral world